

DAXHIEL PERUGORRIA RUCIEL

(469) 673-9154 · daxperugorria@gmail.com · daxhielp.github.io

EDUCATION

University of Texas at Austin – Austin, TX

2025 - 2029

Bachelor of Science in Computer Science

Coursework: Data Structures and Algorithms, Computer Systems, Computer Organization and Architecture, Discrete Mathematics, Linear Algebra, Probability and Statistics

EXPERIENCE

Project Advance Austin – Austin, TX

August 2026 – Current

Software Engineering / Data Analytics Consultant

- **Engineering an interactive geospatial analytics tool** in Tableau, mapping geographic coordinates and relational distances between incarcerated mothers and their children across Texas and the U.S. to drive nonprofit fundraising.
- **Ingested, cleaned, and structured relational database records** of families served, resolving anomalies in geographic, demographic, and facility data.
- **Synthesizing multi-source empirical literature and program metrics** into an online data-driven publication, providing technical documentation to support data-driven grant and corporate sponsorship acquisition.

Code2College – Remote

June 2026 – July-2026

Course Instructor

- **Instructed 24+ students** over an 18-session curriculum, achieving 95% average attendance and 97% average quiz scores.
- **Taught SDE fundamentals**, including Python programming, object-oriented design, error handling, and basic data management.

PROJECTS

Recommendation Engine Evaluation Model | *Datahack 2026, ML*

- **Architected a causal Transformer skip-prediction model** for sequential listening behavior by converting each session into a token stream (user attributes + track semantics + context + labels), enabling next-step prediction of the final skip label and reaching 87.07% validation skip accuracy during training.
- **Scaled representation learning to over 1 million tracks** by generating 3-level hierarchical “semantic” track IDs via MiniBatchKMeans clustering (32×32×32) on standardized audio features, then encoding each track with clustered IDs + binned popularity/skip-rate signals to improve generalization on long-tail content.

VibeDungeon | *AI Agent Game*

- **Designed and built a modular Python RPG engine** with AI-generated encounters and procedurally-generated dungeon maps, along with isolated combat, character, and dungeon systems behind clean interfaces.
- **Integrated an LLM-driven agent layer** by introducing provider-agnostic agent abstractions, resulting in AI-generated narrative content isolated from core game logic.

LEADERSHIP

Austin Spanish First – Austin, TX

2020 - Current

Co-Director

- **Direct a 10+ member technical team**, delegating tasks and coordinating efforts to deliver seamless audio/visual experiences for 400+ weekly attendees.
- **Collaborate with cross-functional teams** to provide a satisfactory viewer experience, occasionally handling logistics for meetings and site setup.
- **Redesigned audio and live-streaming infrastructure**, improving broadcast quality and growing online viewership by 50%.

TECHNICAL SKILLS AND WORK STATUS

- **Tools & Skills:** Claude Code, Data Visualization, Geospatial Analysis, Pytorch, FastAPI, Next.js, React, Vercel, Supabase, Express, NEAT, SQL
- **Languages:** Python, Java, Rust, Javascript, C/C++
- **Work Eligibility:** Eligible to work in the U.S. with no restrictions